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It's Not Just Another Brick in the Wall: The Ceramic Building Materials of Colonia Aelia Capitolina

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ABSTRACT: Following the destruction of Jerusalem in 70 CE, the Roman presence brought with it new ceramic building materials, including bricks, roof tiles, pipes, and tubuli, usually in relation to public buildings, particularly bathhouses. Among the most well-known are the tiles bearing stamps of the Tenth Roman Legion. In the Byzantine Period, such materials were stamped with private Greek names. A group of tiles uncovered in the Wilson's Arch excavations fill the chronological gap between the legionary production and the Byzantine period and display the name of Colonia Aelia Capitolina. These were made from a different ceramic 'recipe' than the legionary tiles, indicating they were produced in a separate workshop and implying a shift in the third century CE from military production (and ownership) of public buildings to municipal-initiated construction projects. These tiles support the theory that the original civic centre of Aelia Capitolina was located directly west of the Temple Mount.

SETTING THE STAGE: CERAMIC BUILDING MATERIALS IN JERUSALEM

Following its destruction in 70 CE, ceramic building materials (henceforth CBM) began to widely appear in Jerusalem, introduced by the Romans as a novel approach to local construction (e.g., McWhirr 1979; Hamari 2011; Mills 2013; Cohen-Weinberger, Levi and Be'eri 2020).¹ These materials, likely related to public buildings, particularly bathhouses, include bricks for pavements and walls, roof tiles, tubuli, pipes, and other objects made from fired clay and used as architectural elements in the ancient world. CBM were relatively cheap to make and were mass-produced. Hence, they provided a readily available and

1 The earliest use of CBM in Judea was as tubuli in the bathhouses in Herodian palaces. Prior to 70 CE, the use of CBM was restricted exclusively to royal bathhouses and was not commonly used or mass produced.

durable product, resilient to intense heat and fluctuating temperatures, and easily replaceable and recyclable, as their very common reuse indicates (Craig 2011: 3; Hamari 2011; Harvey 2013: 3).

CBM were often marked by their manufacturer, by stamping, mould-making, or incising. The earliest production of CBM in the vicinity of Jerusalem is attributed to the military, as noted by the stamps of the Tenth Roman Legion — Legio X Fretensis — on numerous roof tiles. The only known workshop in Jerusalem and its vicinity that produced CBM for a large market was excavated west of the Old City in the area presently occupied by the Jerusalem International Convention Centre (Binyanei Ha'Uma, henceforth JICC) and the Crowne Plaza Hotel (Arubas and Goldfus 2005; Be'eri and Levi 2018; Cohen-Weinberger, Levi and Be'eri 2020). This is one of the largest pottery workshops in the eastern Mediterranean and the only Roman legionary kiln site thus far excavated in the eastern provinces. CBM stamped by the Tenth Roman Legion were found in several excavations in Jerusalem, including the Ophel and the Southwestern Hill (e.g., Aharoni 1962; Barag 1967; Geva 2003; Weksler-Bdolah 2020: 4750 and references therein). Additional evidence for CBM production was found in a kiln associated with a Late Roman-early Byzantine villa in 'En Ya'al (Naḥal Refa'im), Jerusalem (Weksler-Bdolah 2016).

While the exact period in which legionary stamped CBM ceased to be produced is uncertain, there is no doubt that this occurred by the late third century CE, when the unit left Jerusalem. Although their production may have ceased earlier, as may be suggested by the abandonment of the JICC workshop at the turn of the second or early third century CE (Cohen-Weinberger, Levi and Be'eri 2020), they likely continued to circulate until much later. However, by the early Byzantine period, there is an increase in stamped bricks bearing Greek names, indicating the existence of private-owned workshops that produced CBM (Peleg 2003; Avner 2015; Weksler-Bdolah *et al.* 2022).

CBM FROM THE WILSON'S ARCH EXCAVATIONS

The excavations beneath Wilson's Arch, located just north of the Western Wall Plaza at the foot of the Temple Mount (fig. 1), revealed eight archaeological strata spanning the Hasmonean era until the Mamluk period. The most significant finds are dated to the Late Roman period and include a small theatre-like structure located directly beneath the arch, dating to the mid-second century CE (Uziel, Lieberman and Solomon 2019; Regev *et al.* 2020). The various fills and deposits yielded a large assemblage of CBM in the layers dating to the Roman period (Strata 6–4). All CBM from this excavation originated in well-dated and well-defined fills, although in secondary deposition. Therefore, while a secure chronological sequence can be established, the original context remains unclear. The earliest appearance of CBM in the excavations is attributed to Stratum 6, where legionary

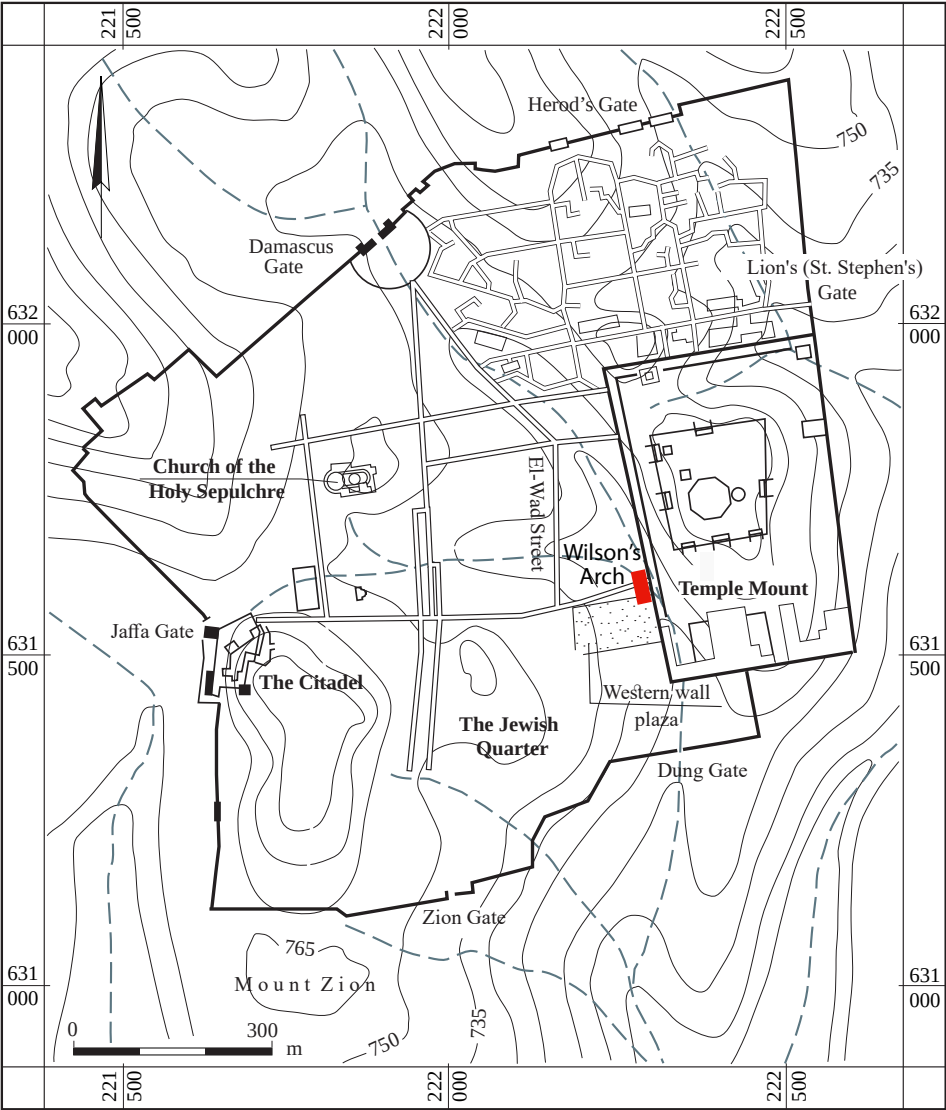


Fig. 1. Location map

stamped tiles were found in the foundations of the theatre-like structure, dating to the early second century CE. Subsequently, in Stratum 5 (late third century CE) and Stratum 4 (fourth century CE), significant amounts of CBM were discovered in layers of stratified fill that sealed the theatre-like structure. These layers differ in character, with the earlier Stratum 5 layer (5a) consisting of a more uniform, thick layer of fill, whereas Stratum 4 was typified by a sequence of alternating sediments, including considerable amounts of dumped ash which yielded large

amounts of charcoal and botanical remains (fig. 2).² The marked CBM retrieved from these strata can be divided into three groups, according to the content of the marking (see table 1 below):

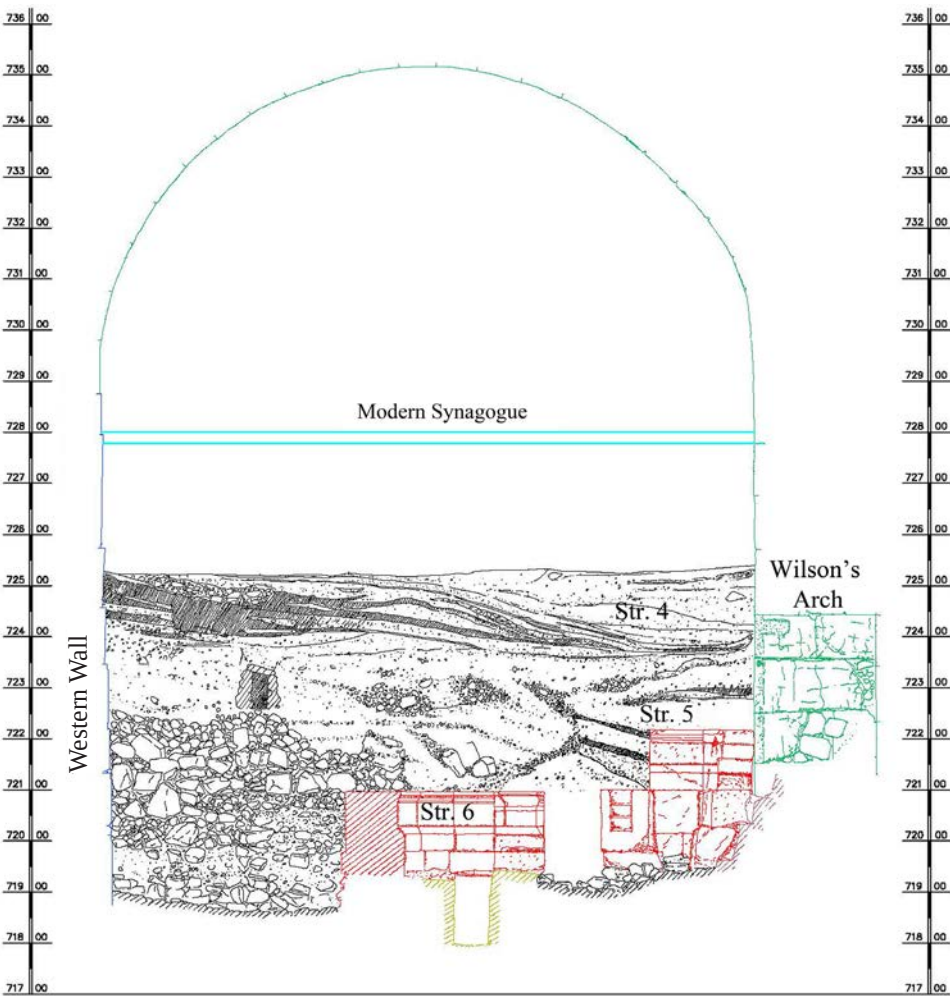


Fig. 2. Section of the excavations beneath Wilson's Arch, showing the relevant strata

2 The botanical remains are currently being analyzed by S. Frumin and E. Weiss of Bar-Ilan University. The charcoal remains are being analyzed by M. Cavanagh and D. Langgut of Tel Aviv University.

1. CBM displaying stamps of the Tenth Roman Legion. These appear throughout Strata 6–4, although it is likely that their deposition in the later layers occurred significantly after their production (see above).
2. CBM marked by the colony of Aelia Capitolina. These appear in Strata 5a and 4. Three distinct sub-types were noted: stamped (18), incised (2), and mould-made (18).
3. CBM stamped in Greek with the name ΠΟΥΠΛΙΟΥ (Πουπλίου) – the Greek version of the Latin name Publius. Two CBM of this type were found in Stratum 4. Thirty such stamped CBM were found in the House of the Menorot in the Temple Mount excavations (Peleg 2003).

Of particular importance is the second group, noting the colony of Aelia Capitolina (figs. 3–4). As opposed to the two other, well-known categories, the Aelia Capitolina CBM are much less common and have gone largely unnoticed. Similar tiles were found in the excavations to the south of the Temple Mount (Mazar 1971: 11, fig. 6:1–3; possibly also Macalister and Duncan 1929: 167–168), in a locus generally dated to the second and third centuries CE, as well as one example in the Western Wall Tunnels (Bahat 2013: pl. 10:13, published with no chronological information) and one in ‘En Ya’al (Avner 2015: 10–12, fig. 8:4) (possibly two, if including the uncertain fragment in fig. 8:3), discovered in a locus containing sherds dating from the Second Temple period to the fifth century CE. Besides these, no other colonial marked CBM have been published from Jerusalem. This is the first time a comprehensive study is carried out on such CBM as a distinct group.

Of the Wilson’s Arch assemblage, 18 fragments, probably all bricks (rather than roof tiles), are marked with rectangular stamps bearing the name of the colony.³ The stamps are read in the genitive denoting ‘(product of) *Colonia Aelia Capitolina*’. Three types of stamps were discovered:

1. Rectangular stamp (table 1:36–37, 49–52); frame width at least 4.2 cm, height 4.2 cm, letters 1.5–2.2 cm; two lines of text, the letter L in line 2 slightly suspended, the letters CA are ligated.
COL(oniae)
AEL(iae) CA(pitolinae)
2. Rectangular stamp (table 1:38–47, 53) bearing a retrograde impression; frame height 4.6 cm, letters 1.2–2.1 cm; the text is the same as No. 1 above.

3 The term ‘brick’ denotes here any flat square tile that could be used in a floor, a wall, or as hypocaust *pilae*, but not for roofing or any other specific function (e.g., tubuli). It is important to note that there is no known building in Roman Palestine entirely constructed with opus testaceum/latericium.

3. Rectangular stamp (table 1: 35, 48) bearing a retrograde impression; frame width 5.1 cm, height 3.6 cm, letters 1.2–1.8 cm:

COL(oniae)

AEL(iae)

Other than these, 15 roof tiles were discovered bearing the name of the colony impressed as part of the tile's mould rather than as a stamp. The letters of these inscriptions are 4–7 cm high and the spaces between the letters are 2–4 cm. It seems that, unlike the small stamps employed for the bricks, the inscriptions on these roof tiles were decorative, as if meant to be seen. Perhaps they were used on rooftops that could be observed from a higher vantage point, which was perhaps, in our case, the bridge supported by Wilson's Arch.

No intact samples were discovered, but the content of the inscriptions is clear and the parsing of the letters can be reconstructed from the comparison of the different tiles. The sequence ELI was discovered on two tiles (table 1:16, 32) and under what seems to be the top line of the tile. In one case, this letter sequence appears at the end of the first line (table 1:32, although this is a trapezoid tile). Two fragments bear a reversed letter S (table 1:20–21); in one instance, the letter appears in a line below the letters OL. We can thus suggest that the inscription was laid out along the length of the tile in two lines and could read:

COL(oniae) AELI(ae)

C(apitolinae) 2

At least in some variants, a second line bore the reversed S shape symbol. It seems odd that the term *Capitolina* is not mentioned, and perhaps it was and will be discovered in future examples. Two fragments bear the letter [--]G[--] (table 1:17, 19), and one has the letter sequence [--]LE[--]. These can be 'accessory letters', i.e., additional letters unrelated to the main inscription, marked on tiles to assist construction, distribution, or any other functional purpose, or perhaps alluding to the name of the manufacturer (Bardill 2004: 20–21). Whereas the reversed S is probably such an 'accessory', the letters L, E, and G may have formed the inscription LEG(io), referring to an army legion, which in the case of Jerusalem would be the Tenth Legion. If so, such an inscription indicates that the workshop from which the tiles originated serviced both the colony and the legion, and was active before the departure of the latter at the end of the third century CE.

The third group is represented by only two tile fragments with inscriptions incised (not impressed) before firing. One example carries the letters COL (fig. 4:33), similar to the abbreviation that appears on the other groups of CBM. In the second example, the letters LO were incised (table 1:34). LO may be part of the name Colonia, but the reading cannot be confirmed due to the limited corpus. As opposed to the impressed or mould-made groups of CBM which were confined to a rigid formula, incised inscriptions were carved individually into the tiles and

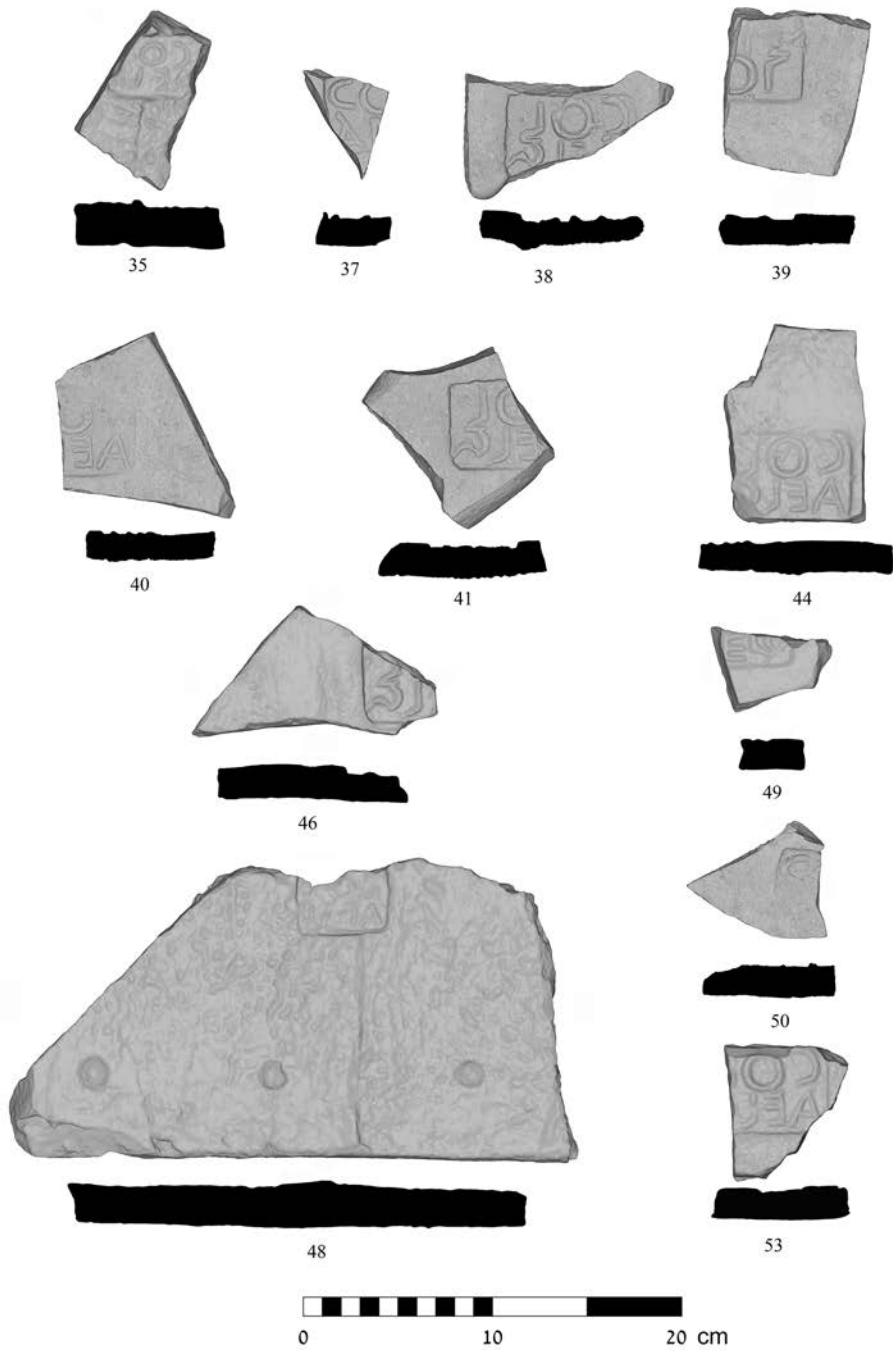


Fig. 3. Stamped CBM of Aelia Capitolina

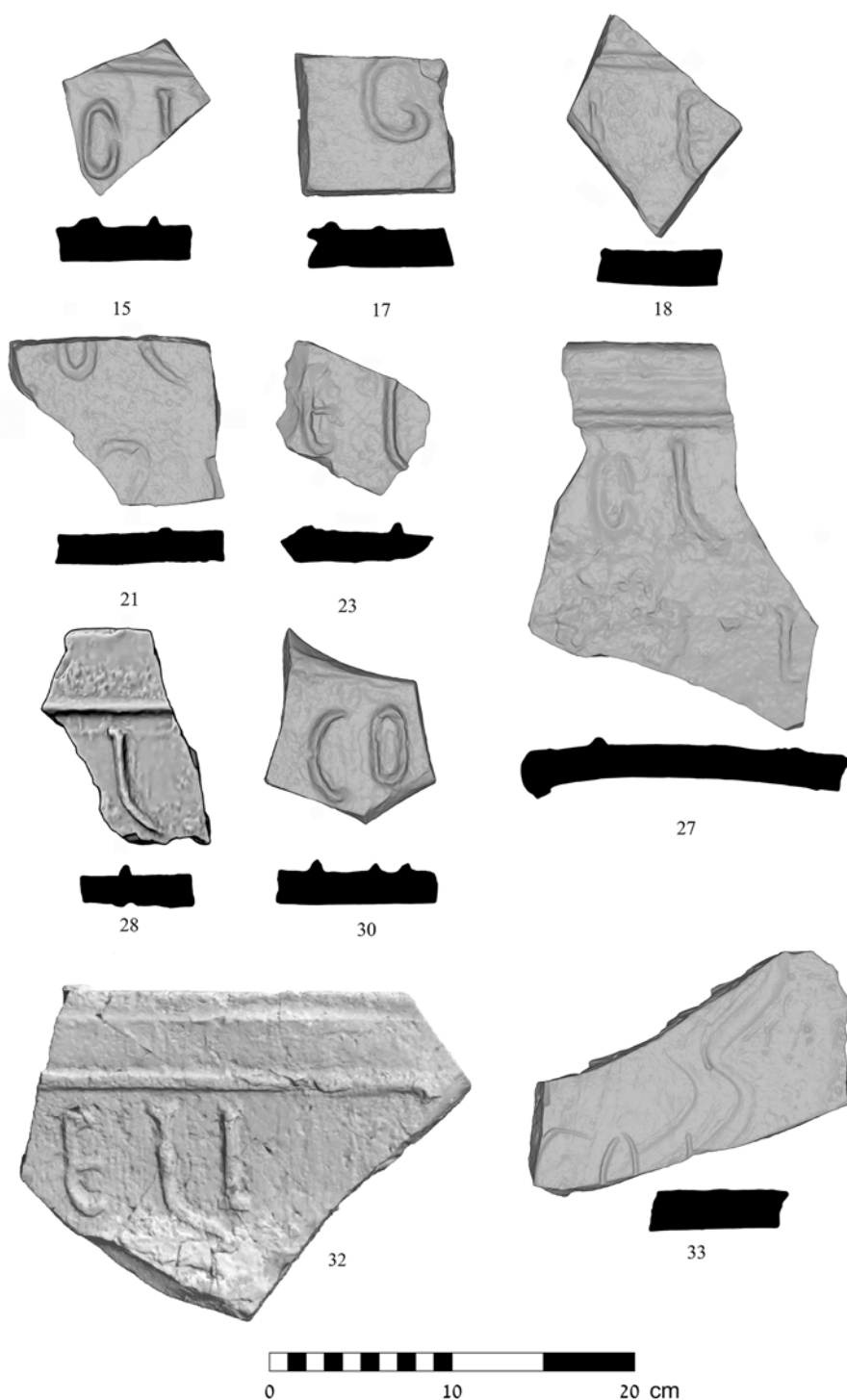


Fig. 4. CBM: mould-made (15, 17–18, 21, 23, 27–28, 30, 32) and incised (33)

in a sense, resemble graffiti writing more than formulated labelling. Thus, incised tiles would likely have carried a much wider variety of inscriptions than stamped or mould-made CBM. In addition, by hand-marking each tile individually, the manufacturing process must have been significantly slower, which might be the reason why only two examples can be associated with this group.

THE PUBLIC PRODUCTION OF CBM OUTSIDE OF JERUSALEM

Most of the known, organized, public production of pottery in the Roman Empire was confined to military workshops, as opposed to private, commercial entrepreneurs. The sparse evidence for municipal or state activity of pottery production was restricted to brick and tile manufacture. Municipal brickworks were controlled and run by town authorities, owned by the council, and leased to brick workers who operated on their behalf. The CBM produced in such a workshop were usually destined for official projects and structures, although the private sector may have used these to a minimal extent (Peacock 1979). Obviously, not every public structure carried markings that could associate its construction with a specific private or public entity. In Rome itself, the earliest municipal control of the brick industry is first known during the time of Diocletian, although evidence from other colonies and cities is earlier (Peacock 1982: 151).

Although not frequent, in several colonies, cities, and provinces throughout the Roman Empire there is evidence of civil public production of bricks and tiles. The military making of such building materials is well known, carrying a variety of impressions including the name, abbreviation, and/or number of the manufacturing legion or auxiliary unit. In places where municipal impressions were discovered, they usually derive from a former military production in the same area. Municipal production did not necessarily replace the military endeavor and occasionally they coexisted for up to a couple of centuries (Darvill and McWhirr 1984: 249). As in the military workshops, the municipality-made tiles and bricks carried an identifying impression marking the name of the city or colony, and under the auspices of which it was produced.

In Corinth, a large number of colonial-stamped tiles were found, published from two main locations in the city: the Odeum and the sanctuary of Demeter and Kore. The most common impression was the abbreviated official title of the Roman colony: COL L IVL COR (Colonia Laus Iulia Corinthiensis). Many variants of this abbreviation were found. Some of the examples, dating to the first century CE, have additional letters signifying the manufacturer, while others had a double stamping: one for the colony and one for the manufacturer. Another group of colonial tiles was also impressed by two stamps: the name of the manufacturer and a stamp bearing the Greek transliteration of the colony's Latin name: ΚΑΙ, short for *κωνεια λανς ιουλια*. These tiles seem to appear in the second century CE when the use of Greek became more common (Broneer 1932: 137–139;

Bookidis and Stroud 1997: 449, 471–472; Millis 2010: 29; Lancaster 2015: 68).

Colonial production of tiles was also noted in the first century CE colony of Knossos, where tiles retrieved from houses and graves at the site had the imprints of the letters CIC (Colonia Iulia Cnosos) accompanied by an icon of the labyrinth. It appears that the production of such tiles was undertaken in a tiler under municipal management (Sackett and Cocking 1992: 409–410).

In and around the colony of Glevum in Roman Britain, the most common impression on tiles reads RPG (Rei Publicae Glevensium) (Darvill and McWhirr 1984: 247–248). Other than the RPG stamp, these tiles often included details of administrative officials of the colony. RPG stamped tiles were initiated during the first two decades of the second century CE and were discovered in the colony, as well as in villas in the nearby vicinity. It has been suggested that the distribution of the tiles reflects the territory of the colony and that most of the structures which contained RPG tiles or bricks were public initiatives, although some were probably reused in private structures or were originally marketed for private buildings.

Provincial stamps from the first–second centuries CE in Britain are also known from London. Tiles with the city’s impression were stamped PR BR LON (Provinciae Britanniae Londoni) and other slightly varied abbreviations. For example, an additional P (procurator) appears occasionally (Betts 1995). Such tiles were mostly incorporated in public buildings, although some were presumed to be sold to private contractors. Manufacturing and use of provincial stamped tiles are linked with a massive, public-building program commenced in London after 70 CE, including administrative buildings, baths, and palaces (Betts 1995: 222).

PETROGRAPHIC ANALYSIS

Comparative Material, Sampling, and Basic Premises

Forty-one CBM from Wilson’s Arch were sampled, including those that were marked (table 1) and unmarked (table 2), in order to define distinct petrographic groups and possibly identify production areas, indicating the manufacture and supply of such materials.⁴ Samples were chosen from each of the different groups: legionary CBM, colonial marked CBM, and CBM with Greek private names. The fragments sampled were randomly selected out of the examples to represent each of the groups. The raw materials were then compared with 49 previously petrographically analysed CBM stamped by the Tenth Roman Legion from other excavations, including 25 samples from Strata IV–III in the Crowne Plaza Hotel excavations (Cohen-Weinberger, Levi and Be’eri 2020: table 3), four found east of JICC (Cohen-Weinberger, Levi and Be’eri 2020: table 4), three from

4 For more details about the petrographic method, see Quinn 2013.

JICC (Goren 2005), seven from the Jewish Quarter (Ben-Shlomo 2012), six from Ramat Raḥel (see Aharoni 1962), and four from excavations on Mount Zion. The petrographic results of the latter two groups of CBM appear here for the first time (see below).

Table 1. Catalogue of marked CBM, with petrographic groups (NS=not sampled)

No.	Locus	Basket	Str.	Building material	Method of marking	Description	Petrographic Group
1	4245	422187	3	Tegula	Stamped	LGXFR	A?
2	4218	420452	4	Brick	Stamped	Legion stamp	A
3	4218	420268	4	Tegula	Stamped	LXF	A
4	4220	421136	4	Tegula	Stamped	XFR	A
5	4343	422045	5a	Tegula	Stamped	X.F	A
6	4406	423234	5a	Tegula	Stamped	LXF	A
7	4527	423870	5b	Tegula	Stamped	Legion stamp	A
8	4460	424207	5b	Tegula	Stamped	LXF with border	A
9	4542	424072	5b	Tegula	Stamped	Legion stamp	A
10	4553	424419	6	Brick	Stamped	LEGXFR	A
11	4561	424524.1	6	Tegula	Stamped	LEGXF	A
12	4561	424524.2	6	Tegula	Stamped	LEGXF	NS
13	4248	420566	4	Tegula	Stamped	ΠΟΥ[Π] [Λ]Ι[ΟΥ]	B
14	4297	421464	4	Tegula	Stamped	ΠΟΥΠ ΛΙΟΥ	NS
15	4252	421137	4	Tegula	Mould-made	[.]OL[--]	B
16	4259	421386	4	Tegula	Mould-made	[--]EL[--]	B
17	4302	421322	4	Tegula	Mould-made	[--]G[--]	B
18	4238	420589	4	Tegula	Mould-made	[--]LE[--]	NS
19	4424	423216	5a	Tegula	Mould-made	[--]G[--]	NS
20	4419	423209	5a	Tegula	Mould-made	Ⲫ	NS
21	4307	423211	4	Tegula	Mould-made	ÇOL[--] [--]Ⲫ[--]	NS
22	4459	423206	0	Tegula	Mould-made	[--]L[--]	NS
23	4238	420518	4	Tegula	Mould-made	[--]EL[--]	NS
24	4252	420852	4	Tegula	Mould-made	ÇOL[--]	NS
25	4332	422830	4	Tegula	Mould-made	Ç[--]	NS
26	4406	423227	5a	Tegula	Mould-made	[--]I[--]	NS
27	4323	421704.1	4	Imbrex	Mould-made	COL[--] [--]+vacat L[--]	B
28	4289	421244	4	Tegula	Mould-made	[--]L[--]	B
29	4406	423202	5a	Tegula	Mould-made	ÇOL[--]	NS

No.	Locus	Basket	Str.	Building material	Method of marking	Description	Petrographic Group
30	4238	420558.1	4	Tegula	Mould-made	CO[--]	NS
31	4238	420558.2	4	Tegula	Mould-made	L	NS
32	4242	420555	4	Tegula	Mould-made	[--]ELI[--][--]	NS
33	4388	423205	5a	Tegula	Incised	C O L	C
34	4388	423231	5a	Tegula	Incised	L O	NS
35	4243	420566/1	4	Brick	Stamped	Reversed impression: C O [.] A E LIA	C
36	4252	421383	4	Brick	Stamped	[...][.]ELÇ[.]	C
37	4271	421382	4	Brick	Stamped	CO[.] AE[...]	C
38	4276	421185	4	Brick	Stamped	Reversed impression: COL[.]ELCA	C
39	4276	421192	4	Brick	Stamped	Reversed impression: [.]OL [.]LCA	C
40	4349	421867	5a	Brick	Stamped	Reversed impression: [.]O[.] AE[...]	C
41	4243	421375	4	Brick	Stamped	Reversed impression: [.]OL[.]ELCA	C
42	4439	423214	5a	Brick	Stamped	Reversed impression: CO[.] AE[...]	C
43	4256	420839	4	Brick	Stamped	Reversed impression: ÇO[.] AE[...]	NS
44	4406	423229	5a	Brick	Stamped	Reversed impression: COL AELCA	NS
45	4256	421374	4	Brick	Stamped	Reversed impression: [..]L [..]LCA	NS
46	4453	423219	5a	Brick	Stamped	Reversed impression: [..]L [..]LCA	NS
47	4272	421449	3	Brick	Stamped	Reversed impression: [...] AELCA	NS
48	4255	420778	4	Brick	Stamped	Reversed impression: COL AELIA	NS
49	4376	423220	4	Brick	Stamped	[...] [.] E LÇA	NS
50	4248	421139	4	Brick	Stamped	C[.][.....]	NS
51	4276	421208	4	Brick	Stamped	COL AEL[.]	NS
52	4314	421989	4	Brick	Stamped	ÇOL [.]ELCA	C2
53	4345	423215	5a	Brick	Stamped	Reversed impression: COL AEL[.]	NS

Table 2: Petrographically analyzed unmarked CBM

No.	Locus	Basket	Str.	Type	Description	Petrographic Group
1	4406	423200	5a	Roof tile	Unimpressed	A
2	4218	420141	4	Tubuli		A?
3	4419	423201	5a	Brick		A
4	4494	423998	5b			A
5	4457	423126	5a	Roof tile		A
6	4248	420761	4			A
7	4243	420869	4	Roof tile		B
8	4263	421389	4	Roof tile tegula	Unimpressed, red slip	C
9	4321	421394	4	Roof tile tegula	Incised wavy line	C
10	4332	421515	4	Thin roof tile	Unimpressed	C
11	4345	422350	5a	Roof tile tegula	Unimpressed	C
12	4445	423218	5a	Roof tile	With wavy line	B/C
13	4332	421508	4	Tubuli		C
14	4314	421989	4	Roof tile		C (variant)

Petrographic study of ceramic vessels and CBM from the legionary workshops in the JICC (Cohen-Weinberger, Levi and Be’eri 2020) illustrated a significant change after 70 CE,⁵ with new pottery types (e.g., dolia, mortaria) and building materials (roof tiles, bricks, and tubuli) produced using a new ‘recipe’ based on a different geological unit. This change included the intentional addition of quartz grains to the paste in order to significantly increase the toughness of the products (Cohen-Weinberger, Levi and Be’eri 2020). This unique ‘recipe’ helps in attributing vessels and building materials from other sites in the vicinity to these workshops. Additional comparisons were made to two other groups of CBM tiles known from Jerusalem in order to possibly identify similarity to the studied group. These include a tile from the ‘En Ya’al kiln (Cohen-Weinberger 2016) and CBM stamped with Greek private names from the Ophel (Cohen-Weinberger 2003).

5 Before 70 CE, a ceramic workshop operated in the area of the Crowne Plaza Hotel next to the JICC. After the 70 CE destruction, the Tenth Roman Legion took control of the workshop, possibly employing the same Jewish craftsmen, at least during the first phase of its Roman existence (Be’eri and Levi 2018: 32). During the early second century CE, these kilns ceased to be used and the Tenth Legion established the legionary workshop.

Results of the Petrographic Analysis

The CBM were petrographically analysed and divided into three petrographic groups.

Group A (17 samples): This group is characterized by an optically active and rather calcareous matrix. The sand-sized non-plastic components comprise ~10–20% of the paste and contain abundant, poorly sorted, fossilized mollusc shell fragments and coarse (up to 1 mm), rounded, and commonly equant quartz grains. The molluscs often have an infill of iron oxides. A few sand-sized rhombohedral dolomite crystals and dolostone fragments, and a few rare echinoids are also present (fig. 5:1–2). All the legionary CBM belong to this group (table 1:1–11).

Group B (seven samples): The matrix of this group is optically active and rich in tiny (<50 µm) rhombohedral dolomite crystals which comprise ~30% of the

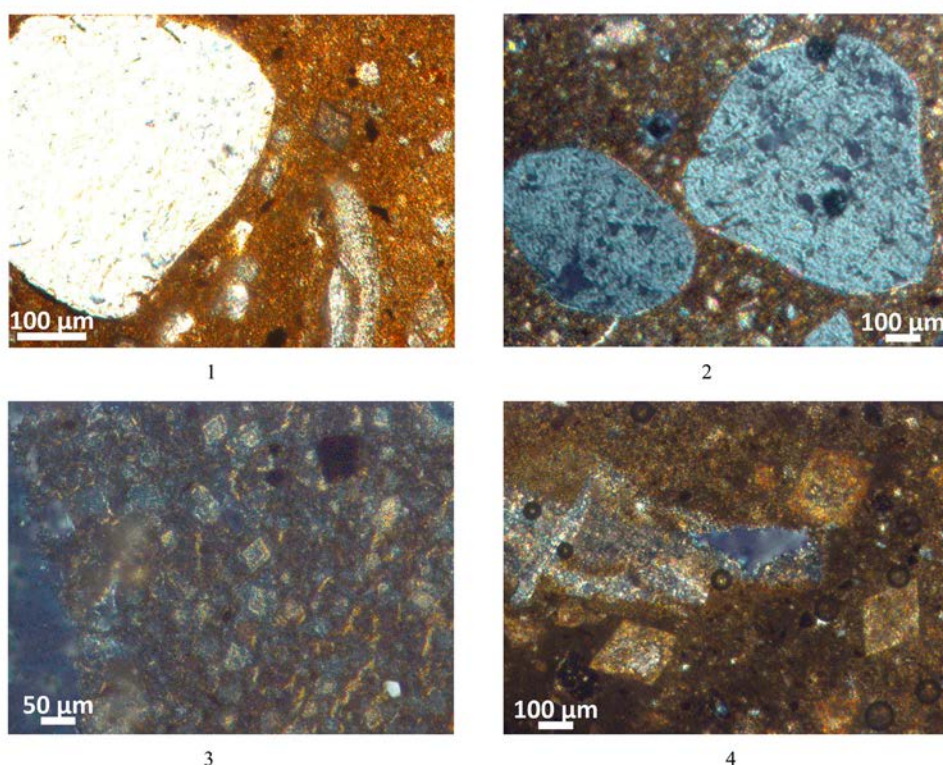


Fig. 5. Photomicrographs of CBM: **1.** Coarse quartz grain, rhombohedral dolomite crystal, and mollusk fragments embedded in calcareous matrix, Petrographic Group A (table 1:5), ppl; **2.** Coarse rounded quartz grains embedded in calcareous matrix, Petrographic Group A (table 1:4), xpl; **3.** Silt-sized rhombohedral dolomite crystal, Petrographic Group B (table 1:28), xpl; **4.** Fine sand-sized rhombohedral dolomite crystal and mollusk fragments embedded in matrix, Petrographic Group C (table 1:35), xpl

paste (fig. 5:3). It contains nodules of ore minerals as well as rarely optically active clay nodules or shales. The sand-sized non-plastic components comprise 10% of the paste and contain mainly rhombohedral dolomite crystals and a few mollusc shells, nari, and dolostone fragments. Rounded ferruginous silty nodules of terra rossa soil are observed in two samples (e.g., table 1:13, table 2:7). In No. 13, the rhombohedral silt-sized dolomite crystals were poorly preserved, as noted in the example from the Temple Mount (see above and Cohen-Weinberger 2003). All mould-made Aelia Capitolina CBM (table 1:15–16, 27–28), the single mould-made [--]G[--] inscription (table 1:17), and the stamped Greek inscription (table 1:13) belong to this group.

Group C (16 samples): This group is characterized by an optically active matrix with up to ~5% silt-sized quartz grains. The matrix contains optically active clay nodules or shales. The sand-sized non-plastic components (up to ~10% of the paste) contain rhombohedral dolomite crystals unevenly distributed, a few mollusc shells fragments, and rarely, quartz geodes (fig. 5:4). Occasionally, nodules of terra rossa soil containing silt-size quartz grains appear in the clay. The matrix of a single sample (table 2:14) is somewhat ferruginous silty and may represent a variation of this group. All the stamped Aelia Capitolina CBM belong to this group (table 1:33, 35–42, 52).

The three different petrographic groups are characterized by different matrices. The matrix of Group A is rather calcareous, the Group B matrix is rich in tiny (silt-sized) dolomite crystals that are absent in Groups A and C, and Group C is optically active and more clayey in nature than Group A. Sand-sized rhombohedral dolomite crystals appear in all groups, but are less frequent in Group A. Abundant fossilized mollusc shell fragments appear in Group A and are much less frequent in Groups B and C. The coarse quartz grains appear only in Group A.

Interpretation of the Petrographic Results

Groups A–C derive from the Moza Formation and are local to the region. The Moza Formation is exposed over vast areas on the Judaeen–Samaritan mountains (Sneh *et al.* 1998; Sneh and Avni 2011) and was extensively used for pottery production (e.g., Goren, Finkelstein and Na'aman 2004; Adan-Bayewitz *et al.* 2008). Petrographic studies conducted on pottery from the Jerusalem area highlighted two groups that are derived from the Moza Formation (e.g., Goren 1995; Goren, Finkelstein and Na'aman 2004). One is clayey in nature (Group C in this study) and the other marly, containing tiny dolomite crystals (Group B in this study). This division follows the geological works of Shalem and others (e.g., Shalem 1925; 1926). Assessment of detailed geological columns of the Moza Formation in different sections of the formation points to a more complex picture and the existence of various clay, marl, limestone, and mollusc-rich layers (e.g., Arkin, Braun and Starinsky 1965: 16–17; Bentor 1966; Rosenfeld and Hirsch 2005).

The raw material of Group A is identified as a marl unit. Previous petrographic studies of the CBM from the Crowne Plaza Hotel excavations indicate that this raw material is derived from a specific geological unit and stratigraphic level of the Moza Formation (Cohen-Weinberger, Levi and Be'eri 2020) that was quarried from a different geological unit and stratigraphic level of the Moza Formation than those used in Groups B and C. The coarse quartz grains most likely originated in the coastal sand dunes of Israel and were deliberately added to the paste. The sand-sized dolomite rhombs most likely originate from the capping Aminadav Formation.

The raw material of Group B is derived from another marl unit of the Moza Formation and the inclusions which make Group A unique are lacking. This group was extensively described in petrographic studies of pottery from excavations in the Judean-Samaritan mountains (e.g., Goren 1995; Cohen-Weinberger and Rosenthal-Heginbottom 2019).⁶

The raw material of Group C was most likely derived from a clayey unit of the Moza Formation, with the sand-sized non-plastic components mainly derived from the capping Aminadav Formation. A single CBM (table 2:14) is a variant of Group C and was made of soil that presumably developed from the Moza Formation. The marls and clay of the Moza Formation are often covered by soil (Arkin, Braun and Starinsky 1965: 16).

Conclusions from the Petrographic Analysis

All analysed CBM bearing stamps of the Tenth Roman Legion were attributed to Petrographic Group A (table 1:1–11). This unique 'recipe' was not used before the legion settled in Jerusalem (Cohen-Weinberger, Levi and Be'eri 2020). Several CBM without impressions also belong to this petrographic group and were presumably produced in the JICC legionary workshop. Not all CBM produced in the Roman workshops were impressed, therefore these CBM could derive from the same workshop and the same chronological phase as the impressed samples, or from an earlier chronological phase of production, dating between 70 CE and the onset of legionary stamped tiles. Alternatively, as the stamp only occupies a small area of the object, such objects may have been stamped, although only the unmarked portion was preserved. Additional studies of legionary stamped tiles from other excavations in and around Jerusalem were also attributed to Petrographic Group A, such as the sampled objects from Ramat Raḥel and Mount Zion. It appears that this ceramic 'recipe' ceased to be used when the JICC workshops were no longer active around the early third century CE (Cohen-Weinberger, Levi and Be'eri 2020). Despite the ceased production, legionary

6 This petrographic group consisted of the prevailing raw material for the post-70 CE fine tableware from the Western Wall Plaza Excavations (Cohen-Weinberger and Rosenthal-Heginbottom 2019).

JICC-made CBM were in use successively for many years (e.g., Ben-Ami and Tchekhanovets 2010).

The Aelia Capitolina markings do not belong to Group A, clearly indicating they were not produced at the JICC kilns. This may be of chronological importance, suggesting that they were only produced in the third century CE, after the workshops ceased to exist. This is further supported by the findspot of these objects, lacking in both Strata 6 and 5b, likely indicating a date of appearance in the second half of the third century CE (see further discussion below).

The mould-made CBM all belong to Petrographic Group B and are similar to the fabric of a CBM stamped ΠΟΥΠΑΙΟΥ (table 1:13), as well as to a tile bearing the same name from the Temple Mount excavations (Cohen-Weinberger 2003; Peleg 2003). It is also similar to a roof tile from the 'En Ya'al kiln (Cohen-Weinberger 2016).

All the stamped and incised Aelia Capitolina artifacts were made from Group C, indicating they were produced in a workshop/s in the vicinity of Jerusalem different than the legionary materials (Group A), the mould-made Aelia Capitolina CBM, and the CBM bearing private Greek names, the latter two made from the same raw material (Group B).

DISCUSSION

The large assemblage of marked tiles from Wilson's Arch produced under the auspices of the colony of Aelia Capitolina has significant implications for our understanding of CBM production, the position of the colony in undertaking building activities, the relationship between military and civilian production, and the location of the colony's civic centre. The production of colonial tiles could not predate 130 CE, when the colony was established, or post-date the reinstatement of Jerusalem in the early fourth century CE. This chronological framework may be narrowed down further, as the use of Latin became increasingly uncommon in the fourth century CE. Furthermore, the presence of the colonial marked tiles beginning only in Stratum 5a at Wilson's Arch (dated to the second half of the third century CE), with none discovered in association with the founding phase of the theatre-like structure, may suggest that they were not produced before the third century CE. This is further supported by the difference in fabric between the Aelia Capitolina CBM and the legionary objects produced in the JICC kilns, which went out of use by the early third century CE. Yet, legionary and civilian production do not necessarily contradict each other and could occur simultaneously, as in Roman Britain. That said, the secondary context in which the colonial CBM were found precludes a firm chronological conclusion.

Of great importance is the petrographic analysis, which clearly distinguished between the colonial and the legionary tiles produced in the JICC workshop. Although it is not possible to determine whether these tiles were produced

simultaneously or subsequently, they were clearly not made by the same craftsmen in the same workshop. Since petrographic analysis carried out on dozens of JICC CBM samples implies a homogeneous composition was used for their fabrication, the possibility that a second petrographic composition, or clay 'recipe', was used in the same workshop is not viable. Furthermore, no colonial marked CBM were discovered in the JICC excavations, supporting the assumption they were manufactured in a different venue rather than in the same workshop that changed its fabric composition or raw-material source.

It appears likely that the municipal production replaced the military production, with a possible overlap, as the settlement of Jerusalem shifted towards a more civilian character. It is possible that the shift is an indication of a handover of production of CBM from the military to the municipality, and later to private entrepreneurs who took over CBM production. A similar situation is noted elsewhere in the empire, such as in Holt, Dalmatia, and Chester, where the Twentieth Roman Legion sublet, leased, and subcontracted production facilities with private contractors (Swan and Philpott 2000; Warry 2010). It was similarly suggested that the legionary tiler at the JICC workshop was leased or sold to a private manufacturer named Iulius who was a veteran, previously employed by the legion as a craftsman (Cohen-Weinberger, Lieberman and Szanton 2022).

Most of the tiles with a mould-made inscription carry a colonial marking (Group B). However, three examples (table 1:17–19) may suggest that the same tiler, as implied by the similar petrographic composition and the same method of mould-made marking, provided its services to the legion as well. Currently, these tiles cannot be dated earlier than the third century CE, and not later than the end of that century when the legion departed. During this period, after the legion was no longer producing its own CBM in the JICC workshop, they were presumably supplied with CBM by other manufacturers. The presence of colonial marked tiles alongside legionary tiles indicates that at least at a certain point, after the JICC kilnworks discontinued production, both the colonial authority and the military legion were initiating construction in Jerusalem. While during this period Jerusalem was already officially recognized as a civilian colony, apparently the legion was still a dominant entity leading construction projects in the colony.

Tiles with public-entity marking are known from other places in the Roman Empire, although it was not a very common practice. Municipal marking of CBM indicates they were produced for the sake of the colony or perhaps under the auspices or supervision of the municipal authority. Such tiles were normally produced in publicly owned workshops which were operated by private workers who leased the workshop from the council and produced tiles for public construction. In the colony, public building was usually the responsibility of the local government through planning, financing, and actual construction (Darvill and McWhirr 1984; Warry 2010: 139; McComish 2012: 50–56).

As in other colonies, such as London, where colonial tiles were produced as

part of a wave of extensive development and construction initiated by the council, a similar situation possibly occurred in Aelia Capitolina. After the destruction of Jerusalem in 70 CE, the city was left in ruins. The Tenth Roman Legion situated their camp in the wreckage of the captured city, initiating some construction attested, among other factors, by the abundance of legionary impressed tiles (including in Wilson Arch Stratum 6, the foundations of the theatre-like structure). Eventually, the army camp developed into an official colony of the Roman Empire, Aelia Capitolina, officially declared in 130 CE. Rebuilding the city as a Roman colony was an ambitious project requiring planning, financial prioritization, and considerable execution. Public facilities and institutions were necessities in the newly established colony, aiming to serve the now civil population inhabiting it. Consequently, intensive construction demanded large quantities of tiles and bricks. The municipal tiler was established at Aelia Capitolina to supply building material for the public initiatives and to satisfy the needs of the expanding colony.

Based on combined evidence implied both by the petrography and the epigraphy, at least two distinct workshops were operating under the auspices of the colony's council. The sponsorship of widespread tile production indicates the local council was closely involved in the architectural development of Aelia Capitolina. Colonial markings are not only a manifestation of authority and power of the provincial government, but also indicate that the colonial construction was carried out according to an organized plan articulated and controlled by the local administration. Having at least two and presumably more production workshops indicates that the demand for colonial tiles was high, attesting to the extensive construction apparently initiated by the colony during a short period. It appears that in its early stages, the construction of Roman Jerusalem's public architecture was under the auspices of the military garrison stationed there. Over time, with the growth of the colony, many of the building projects shifted to the municipality's realm.

The colonial marked CBM from Wilson's Arch comprise the largest collection of such items from Jerusalem published to date. This large, unique concentration indicates that one or more public buildings of the colony were located in the vicinity of Wilson's Arch, initiated by the council of the colony to serve the citizens of Aelia Capitolina. These buildings were not constructed before 130 CE and, for an unknown reason, they were dismantled or repaired no later than the second half of the third century CE. The location of such a building or buildings goes hand in hand with other archaeological evidence of public buildings built to the west of the Temple Mount, as noted in Mazar's excavations, the excavations of the eastern *Cardo*, and most recently, in the excavations beneath Wilson's Arch. These colonial marked tiles add an additional facet to the growing evidence that the civic centre of the Roman colony of Aelia Capitolina was originally located in the stretch of real estate enclosed between the Western Wall of the Temple Mount and the Eastern *Cardo*.

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